

Simulation-Based Inference

Exercise 1

1. Logit-Normal Distribution

Let X be a standard Normal random variable, i.e., $X \sim \mathcal{N}(0,1)$. Define the logit-normal transformation $Y = f(X) = \frac{1}{1+e^{-X}}$.

(a) Show that Y takes values in the interval $(0,1)$. Derive the inverse transformation $X = f^{-1}(Y)$.

(b) Using the change-of-variable formula (Jacobian method), derive the probability density function $p_Y(y)$ for Y .

(c) Using Python (or any programming language), generate 1000 random samples from $X \sim \mathcal{N}(0,1)$ (e.g., using Numpy's standard normal sampling method in Python). Apply the logit-normal transformation to these samples and plot a histogram of the resulting samples $\{Y^{(s)}\}_{s=1}^S$.

(d) Overlay the analytical logit-normal PDF you derived in (b) on the plot in (c) to verify the match. Also, vary the sample size ($S = 10, 100, 1000, 10000$) and visualise how the fit changes.

(e) For each sample size S , compute the MCSE. Create a plot of MCSE versus the sample size and determine the relationship between them.

2. LCS Generators

Using Python (or any programming language),

(a) Do a basic implementation of LCS algorithm for generating random numbers. Find out (web search) the relationship between a , c , and m which leads to full period (i.e. a period of m).

(b) For $m = 2^{32}$, use the values of a and c which lead to a full period and plot the (normalised) histogram for 10000 samples to visually inspect the uniformity.